INCH-POUND

MIL-DTL-83729/1F
10 January 2018
SUPERSEDING
MIL-DTL-83729/1E
01 March 2013

DETAIL SPECIFICATION SHEET

CHOPPERS, ELECTROMECHANICAL, SINGLE POLE, DOUBLE THROW, BREAK BEFORE MAKE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83729.

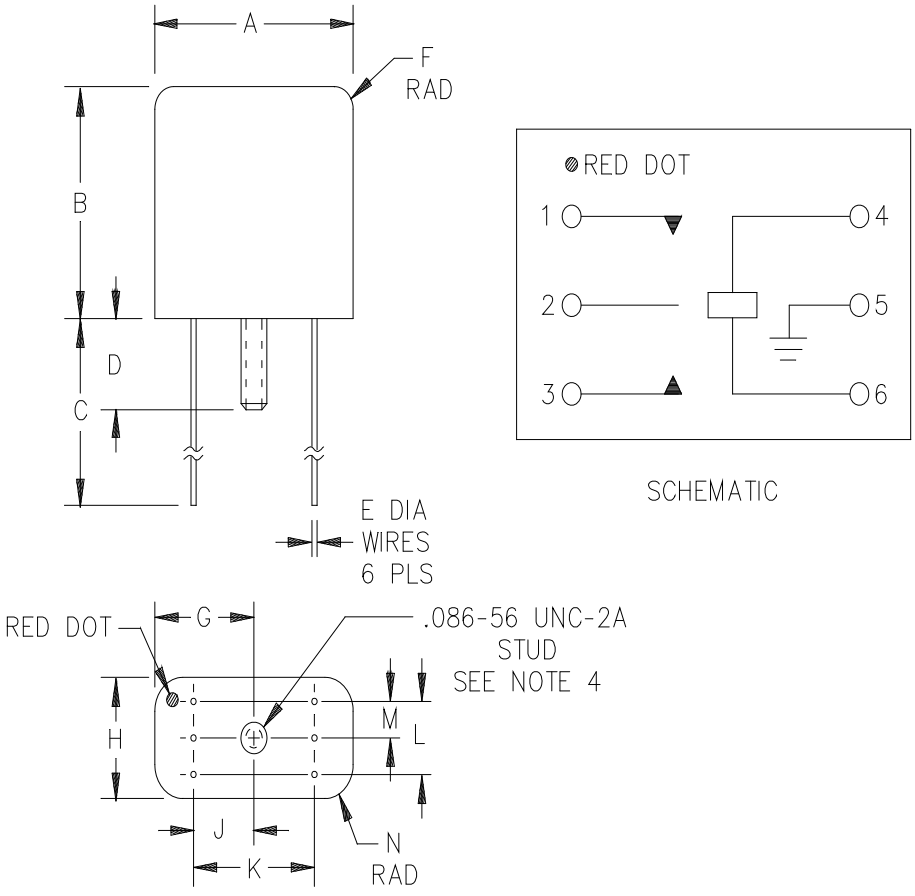

FIGURE 1. Outline drawing, terminal arrangement and schematic diagram.

AMSC N/A

FSC 5945

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Letter	Inches	mm	Letter	Inches	mm
A	.671	17.04	F	.062	1.57
A	.641	16.28	G	.312	7.92
B	.660	16.76	G	.282	7.16
B	.610	15.49	H	.350	8.89
C	1.531	38.89	H	.313	7.95
C	1.469	37.31	J	.200	5.08
D	.265	6.73	K	.400	10.16
D	.235	5.97	L	.200	5.08
E	.017	0.43	M	.100	2.54
			N	.093	2.36

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 (.127 mm).
4. Hardware supplied:
   1 - .086 Flat washer
   1 - .086 Lock washer
   1 - .086 – 56UNC – 2B Hex nut
5. When the coil is de-energized the center reed may be in any of 3 positions.

FIGURE 1. Outline drawing, terminal arrangement and schematic diagram - Continued.

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REQUIREMENTS:

Contact data:

Single pole, double throw, break before make.
Polarity: With positive voltage on lead 6 and negative on lead 4, contacts 2 and 3 close.
Voltage and current: 2 milliamperes resistive maximum at 10 volts, 5,000 ohms resistance.
Resistance: Shall not exceed 10 ohms.
Contact bounce: Shall not exceed 15° during the initial or final 25° of dwell or maximum of 30° for any dwell period.

Coil data at 25°C:

Voltage rated: See table I.
Resistance, dc: See table I.
Impedance: See table I.
Frequency range: See table I.

Waveform characteristics (6.3 volts, at 25°C):

Dwell time: See table I.
Dissymmetry: 30° maximum.
Transfer time: See table I.
Phase angle: See table I.

Noise level: Shall not exceed 30 µV rms.

Environmental conditions:

Temperature range: -65°C to +100° C.
Seal: Seal test I.
Dielectric withstanding voltage: See table I.
Insulation resistance: Shall be 100 megohms minimum at 100 V dc.
Vibration: Vibration I, MIL-STD-202-201, contact derangement 15° maximum.
Moisture resistance: At the completion of the sixth step, insulation resistance shall be 100 ohms minimum.
Shock: MIL-STD-202- 213, test condition I (100 g's).
Dielectric withstanding voltage: Not applicable.

Life:

Duration of test: 1,000 hours at nominal coil voltage and frequency with zero voltage on the contacts at 25°C.

Mechanical:

Weight: .021 pounds maximum, without hardware.
Dimensions: See figure 1.
Dissimilar metals: Not applicable.
Enclosure: Sealed.
Terminals: Wire leads.
Mounting position: Any.

Part or Identifying Number (PIN): M83729/1- (dash number from table I).

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TABLE I. Dash number and characteristics.

Dash Number	Coil Voltage rated	Resistance dc	Impedance	Frequency range	Dwell time	Transfer time	Phase angle	Dielectric Withstanding voltage
001	6.3 volts, 400 Hz at 25°C only	85 ohms ± 10%	115 ohms ± 20%	0 to 1,000 Hz	140° to 185°	2.5° min 400 Hz at 25°C	65° ± 15°	145 Vrms, 400 Hz or 200 Vdc for 60 seconds
002	6.3 volts, 60 Hz at 25°C only	310 ohms ± 10%	330 ohms ± 20%	0 to 250 Hz	155° to 185°	2.5° min 60 Hz at 25°C	25° ± 10°	145 Vrms, 60 Hz or 200 Vdc for 60 seconds

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-DTL-83729, this document references the following:
MIL-STD-202-201 MIL-STD-202-213

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC

(Project 5945-2018-002)

Review activities:
Army – AT, CR4, MI
Navy – AS, MC, OS
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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